Technical Summary: Crop Yield Prediction Analysis

Project: Agriculture / Agritech: Predicting Crop Yield

Author: The Outliers

Date: March 2026

1. Model Overview

The primary objective was to predict global crop yields based on environmental and chemical factors. After evaluating four different algorithms, the **Random Forest Regressor** was selected as the champion model due to its ability to handle non-linear relationships and high-dimensional categorical data (Area and Item).

2. Performance Metrics

The model achieved exceptional accuracy, demonstrating that agricultural yields are highly predictable when outliers are properly managed. The Coefficient of Determination result below was achieved by the model when we stress-tested it using the One-Hot Encoding after it had achieved an R^2 value of 0.9913 when we used Target Encoding.

- **Coefficient of Determination (R^2):** 0.984
- **Mean Absolute Error (MAE):** 3828.83 hg/ha
- **Mean Squared Error (MSE):** 123,434,961.29
- **Root Mean Squared Error (RMSE):** 11,110.13 hg/ha

3. Data Strategy: "The 99% Rule"

The defining technical success of this project was the treatment of statistical outliers. Raw data contained extreme yield values (notably from specialized Belgian farming) that skewed linear models. By **capping yields at the 99th percentile**, we stabilized the variance without losing the "Belgium Signal," leading to a significant jump in model reliability.

4. Top Predictive Drivers

Through feature importance analysis, the following variables were identified as the strongest predictors:

1. **Pesticide Usage:** The strongest correlation to high-intensity yield.
2. **Average Temperature:** Critical for specific crop "Items."
3. **Item Type:** High variance in yield baseline between different crops (e.g., Potatoes vs. Wheat).

5. Conclusion

The model is highly robust and ready for deployment in predictive agricultural planning. Future iterations could benefit from adding soil pH and CO_2 levels to further refine the 1.6% of remaining unexplained variance.